RESEARCH PROPOSAL / 25 OCTOBER 2026

Selective Overlap Records

Preserving task information through ordered neural transitions
and reconstructing it with atlas gluing

Working draft 1.0. Mathematical derivations, implementation proposal, and falsifiable evaluation plan.

1 How we got here

The starting point was a small network that should have been able to solve its task. The QDO study used an eight-dimensional input and a teacher built from four orthogonal features, each rectified separately. A four-unit student already had enough architectural capacity to implement that teacher. Nevertheless, some training runs converged to representations that merged distinctions the answer still depended on. The failure was visible inside the computation: different inputs could reach the same complete intermediate state while requiring different outputs [1, 2].

One especially useful picture was a pair of teacher features A and B whose output coefficients have the same sign. Their input-dependent preactivations need not share a sign. A failed representation could keep their common direction, $A + B$, while weakening their contrast, $A - B$. In one diagnosed cohort, geometric retention was about 99.6% for the common direction and 27.7% for the contrast. These are geometric measurements from that cohort, not percentages of information or of all model error. The distinction mattered because

$$\mathrm{ReLU}(A) + \mathrm{ReLU}(B) \neq \mathrm{ReLU}(A + B)$$

in general. A positive contribution can disappear when it is first combined with a negative contribution and only then rectified. The project also certified a second failure: preactivations that remained different after the linear map became identical after ReLU [1, 2].

QDO approached the problem through optimization. Its historical anchors lived in parameter-derived coordinates, and its repulsive update tried to move training away from configurations already visited. The invariant version removed a per-neuron scaling ambiguity before measuring those distances. That is a different intervention from saving an activation contrast. The manuscript reports a bounded synthetic continuation study; it does not establish a general optimizer or a guarantee of information preservation. In the present research discussion, QDO was judged unsuccessful as the direction to pursue and was set aside. This proposal carries forward the diagnostic lessons, rather than reopening that optimizer [1, 3].

The next intuition was deliberately visual. Imagine arbitrary patterns on a 9×9 grid. Rotate them, place them over one another, and ask what the result still tells us about the original views. Ordinary addition can make different patterns indistinguishable. The suggestion was to make overlapping contributions darker, so that overlap itself leaves a record. The idea quickly acquired two further requirements: a pile may contain more than two layers, and the order of the layers matters. Upper layers should receive different, heavier weights, but those weights must not become another way to erase a lower layer's contribution.

An early mathematical detour treated the requirement of no lost information as exact recovery of every original input bit. That led to positional integer encodings, capacity bounds, and invertible

mixtures of known rotations. Those calculations were useful boundaries: a single ordinary weighted sum cannot encode arbitrary independent real-valued layers, while known rotations of one source are a different problem. They also exposed how floating-point rounding can defeat an algebraic inverse. But complete serialization was a stronger requirement than the developing proposal needed. These results remain in an appendix so that the change of objective is explicit.

The decisive clarification brought back the earlier Hodge atlas work. In the project's description, local charts already have overlaps, transition maps, consistency conditions, and a reconstruction or gluing step. The question became: what must survive between two neural slices so that these local descriptions still determine the target when glued together? A contrast such as $A - B$ is one candidate. The required record can be smaller than a complete copy of the input if every ambiguity it leaves is irrelevant to the required computation [3].

Recurrent Dragon supplied a related discipline in the research discussion: respect causal order, distinguish a useful predictive state from a record of the entire history, and do not infer a preservation guarantee from predictive success. The present design also keeps actual transition data and nonlinear interactions available; it does not substitute scalar coalition attributions for them. Those are design constraints from prior-work discussion, not claims that this proposal has inherited a previously verified implementation [3].

CReLU then provided a concrete comparison. It retains both positive and negative parts of a measured response [4]. The proposed mechanism asks whether only selected contrasts or correction quantities need this protection, with atlas reconstruction aligning them across slices. The proposal is therefore to *capture target-relevant distinctions before they disappear, preserve their layer and chart identities, and glue the retained records into a representation whose remaining ambiguities cannot change the target.*

Central claim to establish. For a declared target or target family T, the complete retained representation E must satisfy

$$E(x) = E(x') \quad \Longrightarrow \quad T(x) = T(x').$$

Overlap compatibility organizes reconstruction. This implication supplies the information-preservation requirement.

2 Proposal, scope, and evidence

The proposed forward computation has an ordinary path and a selective record path. At a boundary where a projection, merge, activation, or later aggregation could discard a target-relevant distinction, a small record is emitted from the information that still exists before that operation. The record carries a layer index and, when needed, a chart or branch identity. Existing atlas maps relate records on overlaps. A reconstruction step combines the records with the ordinary state, and a readout consumes the resulting representation.

The main objective is preservation of a declared computation or task family. Full input reconstruction is a useful reference case, obtained by setting $T(x) = x$, but it is not imposed by default. Exact order recovery is a separate requirement whenever order or source identity must survive independently of the target value. Such metadata remains explicit.

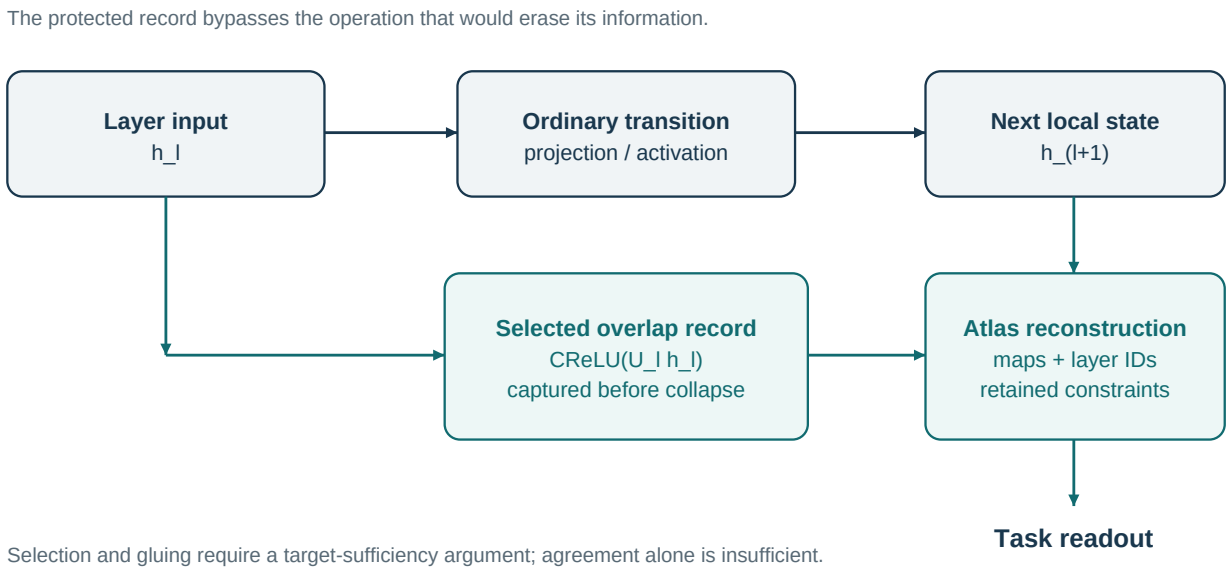


Figure 1: Proposed information path. The record is measured before the destructive transition. Its selected coordinates and the reconstruction both require a target-sufficiency argument. The diagram does not assume that every neural layer is an invertible chart.

Status	What is supported in this proposal
Proved algebra	Target-factorization criterion; exact ReLU cancellation correction; finite merge identities; linear target observability; minimal linear side records; weighted reconstruction and stability bounds.
Checked examples	Deterministic finite-domain checks and explicit negative controls, reproduced by the supplied verification script.
Proposed design	Learned selective records, chart transport, record budgets, and an end-to-end neural readout. These have not been trained here.
Not established	A general learning guarantee, lower total compute, superiority to CReLU or ordinary added capacity, or preservation of every possible future task.

Provenance. QDO and projection-audit statements come from the inspected project manuscript and report [1, 2]. The Hodge atlas mechanism and Recurrent Dragon lessons are used as described in the research discussion [3]; their implementations were not independently re-audited for this proposal. Public literature supplies comparison points, not evidence that the proposed neural system already works. The derivations state their assumptions and make no claim that the elementary identities themselves are new.

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3 The exact preservation criterion

Let $\mathcal{X}$ be the allowed source domain and let

$$T : \mathcal{X} \to \mathcal{Y}$$

be the target, task-feature vector, or specified collection of downstream quantities. Let $E : \mathcal{X} \to \mathcal{E}$ include *everything* retained: ordinary hidden state, side records, layer indices, branch labels, anchors, and any input-dependent transition metadata.

Theorem 3.1 (Target sufficiency)**.** *There exists a decoder $D : E(\mathcal{X}) \to \mathcal{Y}$ satisfying*

$$T = D \circ E \tag{1}$$

if and only if

$$E(x) = E(x') \implies T(x) = T(x') \qquad \text{for all } x, x' \in \mathcal{X}. \tag{2}$$

Proof. If $T = D \circ E$, equal encodings have equal decoded targets. Conversely, for each $e \in E(\mathcal{X})$, choose any x with $E(x) = e$ and define $D(e) = T(x)$. Condition (2) makes this definition independent of the chosen preimage. □

The encoder may identify distinct source states, but only when they have the same required target. Every fiber $E^{-1}(e)$ must lie inside a target fiber. Decoder existence is a structural statement; it does not imply that decoding is cheap, stable, continuous, or learnable from finite data.

For an existing retained state h, if a fiber $h^{-1}(z)$ contains N_z distinct required target values, an added record must have at least N_z distinguishable states on that fiber. For a finite digital alphabet, this requires at least $\lceil \log_2 N_z \rceil$ fixed-width record bits on that fiber. The needed capacity depends on missing target distinctions, rather than automatically on every missing input distinction.

3.1 Task families and unknown future use

For a family $\mathcal{F}$ of future readouts, require

$$E(x) = E(x') \Longrightarrow f(x) = f(x') \quad \text{for every } f \in \mathcal{F}.$$

For a finite family, take $T = (f_1, \ldots, f_q)$. If every possible source distinction matters to some allowed future task, the requirement becomes full injectivity. A selective guarantee therefore needs a declared task family. Keeping a currently known scalar answer can be much easier than keeping the computation reusable for other answers.

For a streaming state with unknown future inputs, two histories may be merged only if every allowed future continuation produces the same required outputs. Agreement on the current prefix's answer alone does not establish this.

3.2 The distributional counterpart

For a deterministic square-integrable scalar target $Y = T(X)$, the best squared-error decoder from $E(X)$ is the conditional mean. Its risk is

$$R_E = \mathbb{E}[\mathrm{Var}(Y \mid E(X))]\,. \tag{3}$$

For any decoder $\widehat{Y} = g(E(X))$,

$$\mathbb{E}[(Y - \widehat{Y})^2] = R_E + \mathbb{E}\Big[\big(\mathbb{E}[Y \mid E(X)] - g(E(X))\big)^2\Big]. \tag{4}$$

Expand the squared error around the conditional mean. The cross term has conditional expectation zero, proving the identity.

Thus $R_E = 0$ gives exact target determination almost surely under the source distribution. It is not a guarantee on inputs outside that support or on measure-zero exceptions. A finite set of successful reconstructions is weaker. For intrinsically noisy labels, preserving the complete predictive law requires preserving the conditional target distribution. If the declared objective is squared error alone, preserving the conditional mean suffices to attain the full-information Bayes risk. Zero conditional variance would instead incorrectly demand removal of irreducible label noise.

3.3 Sequential preservation

Let $s_\ell = (h_\ell, m_\ell)$ be the complete carried state, including earlier records. A boundary update may have the form

$$s_{\ell+1} = G_\ell(s_\ell) = \big(F_\ell(h_\ell), m_\ell, r_\ell(s_\ell)\big).$$

Proposition 3.2 (Composition)**.** *If $T_0 = T$ and, on reachable states, each stage admits $T_{\ell+1}$ with*

$$T_\ell = T_{\ell+1} \circ G_\ell,$$

then

$$T = T_L \circ G_{L-1} \circ \cdots \circ G_0.$$

Proof. Substitute the stage factorizations successively. □

If earlier records are compressed or removed, that operation is another stage to which the same condition applies. A local record can be sufficient together with other records without being sufficient by itself. The complete carried state, rather than an isolated side channel, is the object to audit.

4 Exact overlap corrections for rectified sums

The first constructive case turns the overlapping-shadow idea into an exact local record. A merge can preserve a common component while removing a contrast that still matters to the target. The proposal is to identify that missing contribution before the merge and make it available where the adjacent computation needs it. This formalizes the selective interpretation developed in the discussion [3]. The results below concern specified functions and real arithmetic; they do not assume that a learned selector already knows which features to preserve.

4.1 Projection loss and rectification loss are different

Write $\rho(t) = \mathrm{ReLU}(t) = \max(t, 0)$. A linear projection P loses every distinction in its kernel: $P(x + v) = Px$ whenever $Pv = 0$. Rectification can then lose further distinctions because all nonpositive preactivations produce zero. These are separate operations, with separate requirements for a repair.

The CReLU construction retains the positive and negative parts of a measurement [4]:

$$\mathrm{CReLU}(z) = \begin{pmatrix} \rho(z) \\ \rho(-z) \end{pmatrix}, \qquad z = \rho(z) - \rho(-z). \tag{5}$$

It therefore preserves the supplied measurement z. If $z = Px$, the identity does not recover a component of x already removed by P.

In the motivating pair collapse, let

$$S = A + B, \qquad D = A - B, \qquad H = \rho(S), \qquad F = \rho(A) + \rho(B). \tag{6}$$

Here D is a contrast lost by the sum. The inputs $(1, -1)$ and $(0, 0)$ both have $S = 0$, so even $\mathrm{CReLU}(S)$ gives the same record. Their required targets are nevertheless 1 and 0. This is the distinction identified in the earlier projection example [2]. A record intended to preserve it must branch before the sum removes it. Merely branching before the subsequent ReLU is too late.

4.2 The cancellation correction

Proposition 4.1 (Exact local correction)**.** *For every $A, B \in \mathbb{R}$, define*

$$\kappa(A, B) = \rho(A) + \rho(B) - \rho(A + B). \tag{7}$$

Then $F = H + \kappa(A, B)$, and the following formulas are equivalent:

$$\kappa(A, B) = \frac{|A| + |B| - |A + B|}{2} \tag{8}$$

$$= \frac{1}{2}\rho(|D| - |S|) \tag{9}$$

$$= \begin{cases} 0, & AB \geq 0, \\ \min(|A|, |B|), & AB < 0. \end{cases} \tag{10}$$

In particular, κ is nonnegative.

Proof. Use $\rho(t) = (t + |t|)/2$ in (7). The linear terms cancel, giving (8); the triangle inequality gives nonnegativity. If A, B have the same sign, $|S| = |A| + |B| \geq |D|$. If their signs differ, $|D| = |A| + |B| \geq |S|$. Hence

$$|A| + |B| = \max(|S|, |D|),$$

which yields (9). In the opposite-sign case, $|S| = \big||A| - |B|\big|$, so

$$\kappa = \frac{|A| + |B| - \big||A| - |B|\big|}{2} = \min(|A|, |B|).$$

The same-sign case gives zero, including either input being zero. □

The gate identifies precisely when the collapsed sum loses a contribution needed by this target. A binary gate alone is insufficient: the missing amount varies. The record κ supplies that amount. Conditional on the retained value H, the transformations $F = H + \kappa$ and $\kappa = F - H$ are inverse. Thus this record contains exactly the missing target value conditional on H, without specifying which input supplied the positive contribution or reconstructing the original pair.

Keeping a signed contrast alongside a rectified sum is a different proposal, and does not suffice in general. The following collision has strictly negative sums:

A	B	S	H	D	F
1	-4	-3	0	5	1
2	-3	-1	0	5	2

The retained pair (H, D) is identical while the target differs. Retaining (S, D) would recover $A = (S + D)/2$ and $B = (S - D)/2$. For this symmetric target, $(S, |D|)$ also suffices. The correction (H, κ) provides a more direct statement of what the target requires. None of these facts establishes a computational advantage for one implementation.

The distinction between a gate and a record can also be checked on a whole family of inputs. Set $(A, B) = (t, -t)$ for $0 < t < 1$. Every member has $H = 0$ and the same opposite-sign gate, but its target and correction are both t. An exact record must distinguish these target values; a flag saying that cancellation occurred cannot do so. Conversely, $(t, -t)$ and $(-t, t)$ have opposite signed contrasts but the same target. This target does not require the record to distinguish them. The example identifies what must vary in the record without demanding preservation of an unnecessary distinction.

4.3 Repairing the original four-feature teacher

In the motivating QDO setup, write the four intrinsic teacher features as $u_1, \ldots, u_4$. The teacher is [1]

$$f_\star(x) = \frac{-\rho(u_1) + \rho(u_2) - \rho(u_3) + \rho(u_4)}{2}. \tag{11}$$

Group the features by their output coefficient sign. Define

$$S_+ = u_2 + u_4, \qquad \kappa_+ = \kappa(u_2, u_4), \tag{12}$$

$$S_- = u_1 + u_3, \qquad \kappa_- = \kappa(u_1, u_3). \tag{13}$$

Applying Proposition 4.1 to each pair gives the exact repair

$$\boxed{f_\star(x) = \underbrace{\frac{\rho(S_+) - \rho(S_-)}{2}}_{f_{\mathrm{pair}}(x)} + \frac{\kappa_+ - \kappa_-}{2}.} \tag{14}$$

Indeed, the positive pair contributes $\rho(S_+) + \kappa_+$ and the negative pair contributes $-\rho(S_-) - \kappa_-$. Adding them and dividing by two reproduces every term of (11). Each local record is nonnegative; their contribution to the scalar output has the teacher's signed coefficients.

This construction identifies sufficient information for the collapsed-pair example without requiring recovery of all eight ambient input coordinates. It is an oracle construction when the teacher features u_j are supplied. A student that has not learned those features cannot calculate these exact records from an already collapsed representation. Accordingly, an oracle repair experiment establishes sufficiency of the proposed information; learning an adequate record remains a separate empirical question. Supplying the teacher features to a record branch must be disclosed as additional information in any comparison.

4.4 Position weights and signed readouts

Suppose the required pair target assigns known positive position weights:

$$F_\alpha = \alpha_0 \rho(A) + \alpha_1 \rho(B), \qquad 0 < \alpha_0 < \alpha_1.$$

The later or upper position receives the larger weight. Positive homogeneity, $\rho(\alpha A) = \alpha\rho(A)$ for $\alpha > 0$, allows the substitutions $u = \alpha_0 A$ and $v = \alpha_1 B$. Therefore

$$F_\alpha = \rho(\alpha_0 A + \alpha_1 B) + \kappa(\alpha_0 A, \alpha_1 B). \tag{15}$$

The common/contrast formulas apply to these weighted inputs. The weights belong to positions if moving a layer is intended to change its influence. This target can distinguish some reorderings, but it does not promise to distinguish every ordering of the original data.

Negative output coefficients require care because $\rho(\alpha A) = \alpha\rho(A)$ generally fails for $\alpha < 0$. For arbitrary real α, β, set

$$T = \alpha\rho(A) + \beta\rho(B), \qquad S = \alpha A + \beta B.$$

Expansion of the same ReLU identity yields

$$T = \rho(S) + R_{\alpha,\beta}, \qquad R_{\alpha,\beta} = \frac{\alpha|A| + \beta|B| - |\alpha A + \beta B|}{2}. \tag{16}$$

For example, $\alpha = 1$, $\beta = -1$, $A = 1$, $B = 2$ gives $T = -1$, a zero rectified base, and correction -1. A general signed readout therefore cannot be represented by an unqualified nonnegative correction rule.

4.5 Arbitrary finite piles and exact local merging

Consider a fixed finite collection $u_1, \ldots, u_K$, where positive position weights may already have been absorbed into the u_i. The required target is $F_K = \sum_i \rho(u_i)$. Choose any binary merge tree. Each leaf retains its signed value; each internal node v computes

$$S_v = S_{v,L} + S_{v,R}, \qquad \kappa_v = \kappa(S_{v,L}, S_{v,R}).$$

The child sums must remain signed until this operation has used them.

Theorem 4.2 (Telescoping local corrections)**.** *For every such tree,*

$$F_K = \rho(S_{\text{root}}) + \sum_{v \text{ internal}} \kappa_v. \tag{17}$$

Proof. Expand each internal record as $\kappa_v = \rho(S_{v,L}) + \rho(S_{v,R}) - \rho(S_v)$. Every nonroot internal node appears once as a child, with a positive term, and once as a parent, with a negative term. These terms cancel. The remaining terms are the rectified leaves minus the rectified root, proving the claim. For one leaf the internal sum is empty and the identity also holds. □

The records need not all remain separate when only this aggregate is needed. A subtree can carry a signed sum S and accumulated correction R. Define

$$(S_L, R_L) \star (S_R, R_R) = \big(S_L + S_R,\ R_L + R_R + \kappa(S_L, S_R)\big). \tag{18}$$

Leaves start at $(u_i, 0)$. This gives a constant number of scalar fields; their required numerical range can still grow with the number and weights of the leaves.

Proposition 4.3 (Mass coordinates and the merge law)**.** *Let* $P = \sum_i \rho(u_i)$ *and* $N = \sum_i \rho(-u_i)$*. Then*

$$S = P - N, \quad R = \min(P, N), \qquad P = \rho(S) + R, \quad N = \rho(-S) + R. \tag{19}$$

The merge (18) *is associative and commutative, with identity* $(0, 0)$*.*

Proof. If $P \geq N$, then $S = P - N \geq 0$ and $R = N$, giving both inverse formulas. If $P < N$, then $S < 0$ and $R = P$, again giving them. These formulas establish a bijection between $(P, N) \in \mathbb{R}^2_{\geq 0}$ and $(S, R) \in \mathbb{R} \times \mathbb{R}_{\geq 0}$. Combining two subtrees adds their P and N separately. The resulting correction is

$$\begin{aligned} R_{\text{new}} &= P_L + P_R - \rho(S_L + S_R) \\ &= R_L + R_R + \rho(S_L) + \rho(S_R) - \rho(S_L + S_R), \end{aligned}$$

which is exactly (18). Thus the merge is ordinary addition in (P, N) coordinates. Associativity, commutativity, and the identity follow from addition. □

These algebraic properties hold in real arithmetic. Floating-point implementations can depend on merge order: even the positive leaves $(10^{16}, 1, 1)$ yield different binary64 sums under the two parenthesizations.

One immediate consequence connects the correction directly to rectified sign channels:

$$\boxed{\sum_i \mathrm{CReLU}(u_i) = \mathrm{CReLU}\left(\sum_i u_i\right) + \binom{R}{R}.} \tag{20}$$

The accumulated positive and negative masses differ from rectifying the final signed sum by the same cancellation amount in both channels. The individual seam corrections depend on the merge tree, but their total R does not.

Commutativity is an explicit limit on the record. The final (S, R) does not retain the original order: $(1, -1)$ and $(-1, 1)$ both yield $(0, 1)$. Position weights can make a specified ordered target respond to a permutation; source labels and transition direction must remain available when the subsequent computation requires them. Neither weights nor this two-field summary guarantee arbitrary future queries. In particular, changing the readout weights after aggregation can require distinctions no longer present.

It is useful to distinguish the information needed for the current output from a state that supports a specified later operation. The pair (H, κ) gives the current target, whereas the merge rule deliberately carries the signed S. For example, the subtree states $(S, R) = (-1, 0)$ and $(-3, 0)$ have the same current positive mass, zero, and the same $(\rho(S), R)$. A later request for $\rho(S + 2)$ produces 1 and 0, respectively. The signed sum is necessary for that additional request. If the only request remains the sum of individually rectified leaves, one can instead accumulate their positive mass directly. The appropriate state therefore depends on the operations that will actually follow the overlap. The explicit (S, R) merge is closed under the signed-sum updates defined here; its closure does not extend automatically to a new nonlinear readout, a new set of position weights, or an unspecified future computation.

Signed leaf coefficients. For the more general target $\sum_i a_i \rho(A_i)$, initialize

$$S_i = a_i A_i, \qquad R_i = a_i \rho(A_i) - \rho(a_i A_i) = \begin{cases} 0, & a_i \geq 0, \\ a_i |A_i|, & a_i < 0. \end{cases} \tag{21}$$

The negative case follows by writing $a_i = -c$, $c > 0$, and using $\rho(A_i) + \rho(-A_i) = |A_i|$. At every merge,

$$\rho(S_L + S_R) + R_{\text{new}} = \rho(S_L) + R_L + \rho(S_R) + R_R.$$

Induction therefore recovers the signed target exactly at the root. Here R may be negative. Omitting the leaf corrections would instead compute $\sum_i \rho(a_i A_i)$, a different function.

The aggregate considered here can already be computed by directly summing the rectified leaves. The derivation supplies an exact local accounting mechanism for a collapsed representation, not evidence of a new efficient algorithm. Its architectural value must be tested when records are learned and transported across the actual transitions of interest.

4.6 Numerical and backward-pass qualifications

The local correction itself has a simple stability property. Let $\mathcal{C} = \{(A, B) : AB \geq 0\}$, the union of the two same-sign quadrants. Then

$$\kappa(A, B) = \text{dist}_\infty\big((A, B), \mathcal{C}\big). \tag{22}$$

For opposite signs, reaching $\mathcal{C}$ requires moving at least one coordinate to zero. Moving the smaller-magnitude coordinate attains distance $\min(|A|, |B|)$; either coordinate change is a lower bound for reaching its respective quadrant. Points already in $\mathcal{C}$ have distance zero. This proves the identity. Distance to a fixed set is one-Lipschitz, so perturbing both inputs by at most ε changes the ideal correction by at most ε. This is a property of the function, not a guarantee that every algebraic implementation evaluates it accurately.

Equivalent real formulas need not behave identically in finite precision. For $A = -10^{20}$ and $B = 1$, the mathematical correction is 1. In binary64 arithmetic, (8) can evaluate to zero because the subtractions lose the small difference. The sign-gated minimum in (10) returns 1 for this example. Implement its gate with sign comparisons rather than the product AB, which can overflow. This avoids a particular subtraction error; signed sums, accumulated records, and final reconstruction still require numerical error checks.

Forward identities also need a stated backward convention. Differentiating an exact correction through its source variables gives matching ordinary derivatives away from activation boundaries. Explicitly, when A, B, and S are nonzero,

$$\frac{\partial \kappa}{\partial A} = \mathbf{1}_{A>0} - \mathbf{1}_{S>0}, \qquad \frac{\partial \kappa}{\partial B} = \mathbf{1}_{B>0} - \mathbf{1}_{S>0}. \tag{23}$$

Adding the derivatives of H cancels the common S terms and recovers the two derivatives of F. Thus the exact source-dependent record restores the derivatives of this specified function as well as its values at these points. Detaching the record, replacing it with an approximation, or choosing a different boundary convention can change training. For example, the exact function $\rho(z) - \rho(-z) = z$ has derivative one, but automatic differentiation with $\rho'(0) = 0$ on both branches computes zero at $z = 0$. The absolute-value version of the pair correction can also produce a wrong derivative for the complete target at an internal cancellation boundary. At $A = 1, B = -1$, using $|\cdot|'(0) = \rho'(0) = 0$ gives the automatic derivative $(1/2, -1/2)$ for $H + \kappa$, while the smooth target F has gradient $(1, 0)$. If preservation of a specified training signal is required, that signal and its backward convention must be included in the contract and verified explicitly.

5 Reconstructing targets from overlapping records

The atlas idea becomes precise when compatibility and information preservation are treated as separate requirements. Local records should agree about the features they share. Their combined information should also determine every distinction required by the declared target. Agreement alone does not establish the second property. This section supplies an exact linear model of both requirements, a construction of minimal selective residuals, and a stability bound for reconstruction. These are mathematical components proposed for this architecture; they are not claims that an existing Hodge implementation already provides the desired neural guarantee.

5.1 Coordinate charts and partial views

A genuine coordinate chart $\phi_i : U_i \to V_i$ is invertible on its domain. Two charts describing the same underlying object have the transition map

$$\tau_{ij} = \phi_j \circ \phi_i^{-1} \quad \text{on } \phi_i(U_i \cap U_j). \tag{24}$$

The identities

$$\tau_{ji} = \tau_{ij}^{-1}, \qquad \tau_{jk} \circ \tau_{ij} = \tau_{ik} \tag{25}$$

hold wherever the compositions are defined; the second identity is evaluated on the common triple overlap. For affine transitions $\tau_{ij}(u) = A_{ij}u + b_{ij}$, these conditions imply

$$\begin{aligned} A_{ji}A_{ij} &= I, & b_{ji} &= -A_{ji}b_{ij}, \\ A_{ik} &= A_{jk}A_{ij}, & b_{ik} &= A_{jk}b_{ij} + b_{jk}. \end{aligned} \tag{26}$$

The displayed matrix equalities apply when the affine maps agree on overlaps with nonempty interior in their coordinate spaces. On a smaller constrained overlap, equality is required only on that overlap.

A many-to-one layer is not automatically such a chart. For partial layer views, use observation maps $r_i : \mathcal{Z} \to E_i$ and comparison maps $\rho_{i,ij} : E_i \to O_{ij}$. The common overlap condition is

$$\rho_{i,ij}(r_i(z)) = \rho_{j,ij}(r_j(z)). \tag{27}$$

Neither observation map needs an inverse. Their joint records may still determine the task. Cellular sheaves provide a standard language for compatible local data and restriction maps; their degree-zero global sections are characterized by an agreement operator. The associated spectral theory also studies reducing edge-message dimensions while preserving global sections [5]. The proposal uses this organizational idea without assuming that an arbitrary stack of neural layers is already an atlas or a sheaf with the required observability properties.

Layer identities, overlap domains, branch choices, and directed transition maps belong to the retained representation. Ordered composition in (25) cannot generally be replaced by an unlabeled weighted sum.

5.2 The source map and the agreement operator

Let the source state be $z \in \mathbb{R}^n$, with local records $e_i = M_i z$. Stack them into $e = Mz \in E = \bigoplus_i E_i$. Let $\delta : E \to O$ collect overlap discrepancies, for example $(\delta e)_{ij} = P_{i,ij}e_i - P_{j,ij}e_j$. Correct restrictions satisfy

$$\delta M = 0, \qquad \text{hence} \qquad \operatorname{im} M \subseteq \ker \delta. \tag{28}$$

This says that genuine source records agree. The stronger condition

$$\operatorname{im} M = \ker \delta \tag{29}$$

says that every compatible collection has an explanation within the chosen source model. It is needed for that universal gluing claim, although decoding records already known to equal Mz does not require it.

The ambiguity left about the source is $\ker M$, not $\ker \delta$. The latter contains compatible records themselves. In particular, $\ker(\delta^\top \delta) = \ker \delta$ characterizes agreement; it does not establish that those records determine the original input or target. If (29) holds, compatible records identify a source equivalence class in $\mathbb{R}^n / \ker M$. The remaining question is whether the target is constant on that class.

5.3 An exact criterion for target observability

Write the required linear task features as $t = Cz \in \mathbb{R}^q$. The rows of C may describe several required readouts. Taking $C = I$ recovers the stronger requirement of preserving the entire source.

Theorem 5.1 (Linear target observability)**.** *For unrestricted $z \in \mathbb{R}^n$, the following statements are equivalent:*

1. *Some decoder recovers Cz from Mz for every z.*
2. $\ker M \subseteq \ker C$.
3. $\operatorname{row}(C) \subseteq \operatorname{row}(M)$.
4. *There is a linear matrix D such that $C = DM$.*

Proof. If a decoder exists and $v \in \ker M$, then $Mv = M0$, so it must give $Cv = C0 = 0$. This proves the second statement. Conversely, that statement says that two states with the same records have the same target. In finite dimensional Euclidean spaces, $\operatorname{row}(M) = (\ker M)^\perp$ and likewise for C. Taking orthogonal complements therefore makes the second statement equivalent to the third. Under the third statement, express each row of C as a linear combination of rows of M and collect the coefficients into D. This gives $C = DM$. Finally, $D(Mz) = Cz$ provides the required decoder, completing all implications. □

Corollary 5.2 (A well-defined target for every compatible record)**.** *Assume* $\operatorname{im} M = \ker \delta$*. A linear gluing readout $\Gamma : \ker \delta \to \mathbb{R}^q$ satisfying $\Gamma(Mz) = Cz$ exists if and only if* $\ker M \subseteq \ker C$*. When it exists, one such formula is $\Gamma(e) = CM^\dagger e$.*

Proof. Exactness ensures that every compatible e has a preimage z. Any two preimages differ by a member of $\ker M$. Under the stated kernel inclusion, they have the same target, so assigning $\Gamma(e) = Cz$ is well-defined. Preimages can be added and scaled, which makes this assignment linear. Moreover, $CM^\dagger M = C$ by target observability, proving the formula. Conversely, the zero record has both 0 and every $v \in \ker M$ as preimages. A well-defined readout therefore requires $Cv = 0$ for every such v, establishing necessity. □

Thus individual views can be lossy while their combined records remain exactly task-sufficient. For a restricted source domain $\mathcal{Z}$, the corresponding necessary and sufficient condition is

$$v \in \ker M \cap (\mathcal{Z} - \mathcal{Z}) \quad \Longrightarrow \quad Cv = 0. \tag{30}$$

The unrestricted theorem is a convenient stronger certificate. It is also necessary when the allowed source domain contains an open region: any missing direction can then be scaled into a difference between allowed states.

5.4 The smallest selective linear residual

Suppose Mz is already retained. Additional records $r = Nz$ should supply the target distinctions absent from it. The following result gives their minimum dimension and an explicit decoder. Here $M^\dagger$ denotes the Moore–Penrose pseudoinverse.

Theorem 5.3 (Minimum residual dimension)**.** *The minimum number of additional independent real-valued linear coordinates needed to recover Cz from (Mz, Nz) for every z is*

$$r_* = \operatorname{rank}\begin{bmatrix} M \\ C \end{bmatrix} - \operatorname{rank} M = \operatorname{rank}(CP_K), \qquad P_K = I - M^\dagger M. \tag{31}$$

Choose N with orthonormal rows spanning $\operatorname{row}(CP_K)$. Then N has r_* rows and the exact decoder is

$$Cz = CM^\dagger(Mz) + CN^\top(Nz). \tag{32}$$

For $r_* = 0$, the second term is absent.

Proof. The matrix P_K is the orthogonal projector onto $K = \ker M$. For any successful residual N, its restriction to K must distinguish everything distinguished there by C. The observability theorem, applied on K, gives a factorization of $C|_K$ through $N|_K$. Consequently

$$\operatorname{rank}(CP_K) = \operatorname{rank}(C|_K) \le \operatorname{rank}(N|_K) \le \text{number of rows of } N.$$

This is the lower bound.

For the proposed construction, the rows of N lie in K, so $NP_K = N$. The matrix $N^\top N$ projects onto $\operatorname{row}(CP_K)$, giving $CP_K = CP_K N^\top N = CN^\top N$. Therefore

$$C = C(I - P_K) + CP_K = CM^\dagger M + CN^\top N,$$

which proves the decoder and attains the bound. Finally, $\operatorname{row}([M^\top, C^\top]^\top)$ is the orthogonal direct sum of $\operatorname{row}(M)$ and $\operatorname{row}(CP_K)$: subtracting $CM^\dagger M$ from C produces the second summand, and adding it back recovers the original rows. Their dimensions establish the rank-difference identity. □

This is a precise sense in which residual retention can be selective. For $C = I$, $r_* = n - \operatorname{rank} M$; a smaller task family may need fewer coordinates. The result counts real coordinates, not bits or computational cost. Dense projections may be expensive, and near-zero singular values need a numerical stability analysis even when symbolic rank is full.

The construction also requires a declared target map. If C is known only through examples, a learned estimate of its missing directions does not inherit an all-input guarantee merely by using this formula. One controlled experiment can supply C analytically and verify the construction; a second can learn it and measure the resulting errors. These are different claims. To preserve several linear targets, stack their matrices into C before constructing N. Adding a future target can increase $\operatorname{rank}(CP_K)$ and require another retained coordinate. An apparently unnecessary direction may therefore be dispensable for one fixed readout yet necessary for reusable computation. The selected task family must remain part of the specification rather than an implicit property of the current training batch.

5.5 A three-variable example

Let $z = (a, b, c)^\top$ and retain two adjacent measurements,

$$s_{12} = a + b, \qquad s_{23} = b + c, \qquad M = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}. \tag{33}$$

Its kernel is $\operatorname{span}\{(1, -1, 1)^\top\}$. Individual values are ambiguous, but $a - c = s_{12} - s_{23}$ is exactly preserved. The target $a + c$ changes along the missing direction and is not recoverable. Adding $s_{13} = a + c$ resolves the ambiguity:

$$\begin{aligned} a &= (s_{12} + s_{13} - s_{23})/2, \\ b &= (s_{12} + s_{23} - s_{13})/2, \\ c &= (s_{13} + s_{23} - s_{12})/2. \end{aligned} \tag{34}$$

Known nonzero weights can scale these records without changing their joint kernel. Collapsing them into one weighted scalar is a further operation that requires a new observability check. The example supports combining partial views; it does not assert that arbitrary overlap summaries are sufficient.

For comparison, suppose two slices originally contain (a, b) and (b, c), but their saved overlap records both contain only b. Their agreement operator is $\delta(e_1, e_2) = e_1 - e_2$. Here $\operatorname{im} M = \ker \delta$ holds exactly: every compatible pair is generated by some source. Nevertheless, a and c remain unobserved, and $a - c$ cannot be recovered. This example satisfies the universal compatibility claim while failing target observability. Saving a common intersection is therefore different from saving enough complementary local information to reconstruct the desired global distinction.

5.6 Weighted gluing and target stability

Given observed records y, consider

$$\widehat{z} \in \arg\min_z \frac{1}{2} \left\| W^{1/2}(Mz - y) \right\|_2^2, \qquad W \succ 0. \tag{35}$$

The positive matrix W defines the reconstruction metric. Its block weights can give upper-layer records greater influence. This operation still retains the indexed record vector y and the map M; the scalar objective value is not their encoding.

Proposition 5.4 (Weights and reconstruction error)**.** *Let $H = M^\top W M$, $A = W^{1/2} M$, and*

$$M_W^\dagger = A^\dagger W^{1/2} = (M^\top W M)^\dagger M^\top W. \tag{36}$$

Then $\ker H = \ker M$. For $y = Mz + \varepsilon$, the estimate $\widehat{t} = C M_W^\dagger y$ satisfies

$$\widehat{t} - Cz = -C P_K z + C M_W^\dagger \varepsilon. \tag{37}$$

If $\ker M \subseteq \ker C$, the first term vanishes and

$$\left\| \widehat{t} - Cz \right\|_2 \le \left\| C A^\dagger \right\|_2 \left\| W^{1/2} \varepsilon \right\|_2. \tag{38}$$

Proof. For every v, $v^\top H v = \left\| W^{1/2} M v \right\|_2^2$. Since W is positive definite, this expression vanishes exactly when $Mv = 0$, proving the kernel identity. The minimum Euclidean norm least-squares solution is

$A^\dagger W^{1/2} y$. The identity $A^\dagger = (A^\top A)^\dagger A^\top$ gives (36). Because multiplication by $W^{1/2}$ does not change the row space, $A^\dagger A = M^\dagger M = I - P_K$. Substituting the observation model gives

$$\widehat{t} = C(I - P_K)z + CA^\dagger W^{1/2}\varepsilon,$$

which proves (37). Target observability implies $CP_K = 0$. Applying the operator-norm inequality to the remaining term proves (38). □

Thus positive weights change conditioning and the treatment of inconsistent observations, but cannot restore a missing target direction. Regularization can select a solution from an ambiguous family; it does not identify which member supplied the data. A genuine anchor adds a measurement.

For $r = \operatorname{rank} A > 0$ and a compact singular-value decomposition $A = U_r \Sigma_r V_r^\top$, the target sensitivity factor is

$$\left\| CA^\dagger \right\|_2 = \left\| CV_r \Sigma_r^{-1} \right\|_2 \le \frac{\|C\|_2}{\sigma_{\min}^+(A)}. \tag{39}$$

Small singular values matter through their target coefficients CV_r. If $r = 0$, observability requires $C = 0$, so the recoverable linear target is trivial and the singular-value bound is unnecessary. Moreover, a small gluing residual alone does not bound target error: an error $\varepsilon = Mv$ is consistent with a different source and can change the target by Cv. The noise bound controls the complete record error, not only its inconsistent component. All displayed identities use exact arithmetic; an implementation must also account for rounding and learned-model error.

5.7 Anchors, gauges, and higher-order distinctions

For a connected graph, let B be its oriented incidence matrix and retain differences $d = Bz$. If $Bv = 0$, every adjacent pair has equal values; connectivity then makes v constant. Thus $\ker B = \operatorname{span}\{\mathbf{1}\}$. Differences determine an absolute state only up to an additive offset. They determine a target Cz exactly when $C\mathbf{1} = 0$. A positive weighted total usually fails this test. Retaining one actual node value supplies an anchor; declaring the mean to be zero merely chooses a convention unless the original mean was known. For anchor measurements Gz, the general condition is

$$\ker M \cap \ker G \subseteq \ker C. \tag{40}$$

Disconnected graphs leave one offset per component; vector-valued states leave a vector offset in each component.

Higher-order targets can expose ambiguities invisible to pairwise records. For $x_i \in \{-1, +1\}$, retain all products $r_{12} = x_1 x_2$, $r_{23} = x_2 x_3$, and $r_{13} = x_1 x_3$. Both $(1, 1, 1)$ and $(-1, -1, -1)$ yield the same records and satisfy their compatibility relation $r_{12} r_{23} r_{13} = 1$. Yet the triple product $x_1 x_2 x_3$ changes sign. A sign anchor resolves this particular ambiguity. Compatibility must therefore be tested against the complete target family, including interactions that become relevant later.

5.8 What a Hodge decomposition contributes

For an unweighted finite complex, let B take vertex values to edge differences and K take edge values to oriented face sums. Then $KB = 0$. The edge space has the orthogonal decomposition

$$E = \operatorname{im} B \ \oplus \ \operatorname{im} K^\top \ \oplus \ \mathcal{H}, \qquad \mathcal{H} = \ker K \cap \ker B^\top. \tag{41}$$

Indeed, $KB = 0$ makes the first two images orthogonal, and their joint orthogonal complement is precisely the displayed intersection. Also,

$$\mathcal{H} = \ker(BB^\top + K^\top K), \tag{42}$$

because its quadratic form is $\left\|B^\top e\right\|_2^2 + \|Ke\|_2^2$. Consequently, $\ker K = \operatorname{im} B \oplus \mathcal{H}$: locally curl-free records need not be globally integrable when a harmonic obstruction remains. This distinction, and reconstruction from pairwise differences, is standard in combinatorial Hodge theory [6].

Keeping all three components preserves the edge record. Keeping only the gradient component is a projection. It is justified for exact source recovery when the allowed records are known to lie in $\operatorname{im} B$, or for a specified task when that task ignores every discarded ambiguity. Hodge structure can expose and organize disagreement; it does not establish that curl or harmonic components are irrelevant to a neural target.

The reconstruction contract. Save labeled local records before the required distinctions disappear. Verify their overlap relations, and require that their remaining source ambiguities are target-invisible. In the linear model these are the separate conditions $\delta M = 0$ and $\ker M \subseteq \ker C$; a universal gluing claim additionally needs $\operatorname{im} M = \ker \delta$. Use the target sensitivity bound to assess numerical reliability. These guarantees must be checked again whenever records are merged, compressed, clipped, or discarded.

For nonlinear learned maps, the corresponding global requirement is that equal complete records imply equal targets. A Jacobian kernel test is only local: $x \mapsto x^2$ has nonzero derivative on separate positive and negative intervals while merging their opposite points. Branch identities and cross-region collisions therefore matter. The linear constructions above provide controlled cases and explicit certificates; extending them to the learned nonlinear architecture remains a stated research obligation.

6 A neural implementation to test

6.1 Selected measurements before the first destructive operation

At boundary ℓ, define h_ℓ to be the state *before* the operation that could erase the chosen distinction. The ordinary path is

$$h_{\ell+1} = F_\ell(h_\ell),$$

for example $F_\ell(h_\ell) = \rho(P_\ell h_\ell)$. A candidate protected record is

$$r_\ell = \mathrm{CReLU}(U_\ell h_\ell). \tag{43}$$

The selected signed measurement is recovered by

$$U_\ell h_\ell = (r_\ell)_+ - (r_\ell)_-.$$

A projection with k independent rows creates k signed measurements and $2k$ nonnegative CReLU channels. Count both correctly when comparing activation budgets.

Because the record branches before F_ℓ, it can retain a contrast that the ordinary projection would remove. If $h_\ell = Px$, every subsequent $U_\ell h_\ell$ is identical on inputs having the same Px. It cannot separate an admissible pair that differs in a target-relevant nullspace direction unless another retained

observation already distinguishes that pair. Record placement must precede the first relevant loss, rather than merely preceding the final rectifier.

The retained collection can be written

$$E(x) = (h_L(x), \{(\ell, k_\ell, r_\ell(x)) : \ell \in \mathcal{B}\}),$$

where k_ℓ is a chart or branch identity when needed and $\mathcal{B}$ is the set of protected boundaries. Known shared transition parameters need not be copied into every input record. Input-dependent selection and map choices must be interpretable by the decoder.

This protects selected measurements, not automatically the target. The requirement at a boundary remains

$$(F_\ell(h), r_\ell(h)) = (F_\ell(h'), r_\ell(h')) \Longrightarrow T_\ell(h) = T_\ell(h'),$$

including other persistent records whenever T_ℓ depends on them. An exact correction quantity can sometimes replace selected signed measurements, as in the ReLU example. A learned approximate correction needs its own error analysis.

6.2 An end-to-end linear audit

For a linear chain $h_{\ell+1} = A_\ell h_\ell$, $h_0 = x$, with records $r_\ell = N_\ell h_\ell$, the final observation matrix is

$$\mathcal{M}_L = \begin{bmatrix} A_{L-1} \cdots A_0 \\ N_0 \\ N_1 A_0 \\ \vdots \\ N_{L-1} A_{L-2} \cdots A_0 \end{bmatrix}. \tag{44}$$

For original target $C_0 x$, the complete criterion is

$$\ker \mathcal{M}_L \subseteq \ker C_0.$$

This catches loss at later aggregation even when a record was adequate when first emitted. Nonlinear versions use the global fiber criterion from Theorem 3.1.

6.3 Layer order and heavier upper weights

Layer indices, transition direction, and map domains are part of the object being glued. Composition is ordered. Heavier upper weights can be used as known positive scaling of individual records or as reconstruction weights. Nonzero scaling can be undone while the individual records remain available.

A single weighted scalar total is another compression and requires its own proof. Distinct weights alone are insufficient:

$$1(1) + 2(3) + 3(2) = 13 = 1(2) + 2(1) + 3(3).$$

Even the same values in different orders collide in this example.

The role of weights must be declared. Position labeling, confidence in noisy measurements, and semantic weighting of the target are different operations. The experiments must not alter an unweighted target while claiming merely to improve its encoding. Upper-layer weights can express the intended priority, while the retained indexed records supply the information needed for exact interpretation.

6.4 Learning proposal

A candidate objective is

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{con}} \sum_{e=(i,j)} \|\pi_{i,e}(v_i) - \pi_{j,e}(v_j)\|^2 + \lambda_{\text{cyc}} \sum_{(i,j,k)} \|\tau_{ik}(v_i) - \tau_{jk}(\tau_{ij}(v_i))\|^2 + \lambda_{\text{budget}} \mathcal{B}(U). \tag{45}$$

The cycle term applies only on valid common domains when these transition maps are part of the model. The budget term can penalize retained dimensions or grouped projection rows. This is a proposed training procedure, not a preservation theorem. Learned zero maps can make consistency losses vanish; hard sparsification can delete a necessary record; and a finite penalty need not impose an exact constraint.

Start with a fixed record budget and no input-dependent hard deletion. After the basic mechanism works, test any adaptive gate for target-changing collisions in regions where protection turns off. Oracle teacher directions may be used in explicitly labeled diagnostic controls. Ordinary training should use the declared data and task loss rather than hidden teacher features.

Inference must remain causal: records use current and earlier states. Future-layer supervision can be used in training under a specified protocol, but evaluation must not reveal unavailable future observations. This implements the causal discipline emphasized in the research discussion [3].

6.5 Forward values, gradients, and arithmetic

An exact forward identity implies equal ordinary derivatives wherever the functions are differentiable, provided the computation remains connected. Detaching or approximating a correction can change learning even when some forward values match. At internal activation boundaries, equivalent expressions can produce different automatic derivatives, which need not equal the derivative of the complete function.

For example, $\rho(z) - \rho(-z) = z$ as a real function. If both ReLU branches use derivative zero at $z = 0$, naive differentiation of that graph returns zero there, whereas the identity function has derivative one. A matched backward convention is needed when gradient preservation is part of a claim.

For the pair correction, the absolute-value subtraction can also suffer catastrophic cancellation. At $A = -10^{20}, B = 1$, its mathematical value is one, but the subtraction formula can return zero in binary64. A more stable pair evaluation returns zero when signs match and $\min(|A|, |B|)$ when signs oppose. Compare signs directly rather than forming AB, which may overflow. This improves that local computation; it does not make the complete floating-point pipeline exact.

A local diagnostic

$$\ker DE(s) \subseteq \ker DT(s)$$

is necessary for a differentiable decoder factorization but is not a global proof. Distinct activation regions may collide. The map $x \mapsto x^2$ is locally invertible on separate positive and negative intervals, yet recovering x requires a branch identity. A neural atlas must therefore address cross-region collisions and valid chart domains.

7 Evaluation plan and acceptance criteria

7.1 Stage 0: mathematical contracts

Reproduce the identities and negative controls using the supplied deterministic verification script. Build a small linear prototype with known observation and target matrices, construct a minimum residual basis, and verify its explicit decoder with rational arithmetic. This checks the mathematical implementation; it is not a neural-learning result.

The negative controls are essential. Merged CReLU must fail on a missing contrast. Clipped-sum-plus-contrast must fail on its stated pair. The stated pairwise-product record must fail to determine the triple-product target. Floating-point tests must exhibit the stated rounding and subtraction failures. Favorable examples alone would not distinguish the proposed contract from a weaker one.

7.2 Stage 1: the original small-network diagnosis

Reuse the original teacher family and, where accessible, saved failed checkpoints. First freeze the ordinary representation and train only the selective measurement and readout components. Then run an end-to-end comparison separately. Keep training, model-selection, and final evaluation data distinct, with matched teachers and initializations across variants.

Separate failure at the linear projection from failure at ReLU. An augmented span floor can certify remaining linear information loss. A zero span floor does not certify a hybrid clipped representation; activation-collision checks must follow. The QDO derivation appendix establishes the relevant exact floor and lower-bound distinction.

Use the teacher-aware correction only as an oracle positive control. It demonstrates that a suitable record can repair a chosen failure. It does not establish that a learned model discovers that record from examples.

7.3 Comparators and resource accounting

Comparator	Question it answers
Ordinary ReLU model	What are the unmodified target error and collision patterns?
CReLU on the same projected responses	How much loss comes from sign clipping after the existing measurement?
Ordinary wider model	Is the effect explained by added capacity or compute?
Random protected measurements	Does learned selection outperform a matched random record budget?
Unstructured skip measurements	Does atlas alignment add value beyond carrying the same amount of extra data?
Selected records without learned transport	Are the transition maps responsible for any gain?
Uniform versus increasing layer weights	Does upper-layer weighting help after identity and capacity are held fixed?
Full-rank signed measurement reference	What does a broad, input-preserving reference achieve? Report its larger budget explicitly.

Comparator	Question it answers
Oracle contrast or correction record	Can the diagnosed failure be repaired when the missing quantity is supplied? Keep this separate from learned results.

Match parameter count, activation count, persistent-state bytes, and measured compute as closely as possible. These budgets may not all match simultaneously; report the differences and use separate capacity-matched and compute-matched comparisons where needed. A low-dimensional dense projection can still be computationally expensive.

7.4 Stage 2: multiple boundaries and reusable targets

Test deeper compositions with records emitted before multiple lossy boundaries. Include order-sensitive targets, layer permutations, chart relabeling with correctly transformed maps, delayed dependencies, and mixed-sign coefficients. Include higher-order targets that cannot be determined by the selected pairwise records without an anchor.

Use a declared collection of readouts on the same underlying features to test reuse. A record that stores one scalar answer can perform perfectly on that answer while failing a broader requirement. Distinguish interpolation on the training distribution from the stated admissible domain.

7.5 Stage 3: conditioning and numerical behavior

Perturb records along target-visible singular directions and compare target error with the derived stability bound. Include consistent perturbations $\varepsilon = Mv$: they expose target errors a compatibility residual cannot detect.

Report precision, scaling, overflow handling, and whether correctness means functional tolerance or bitwise equality. Test cancellation near zero and widely separated magnitudes. Check gradient flow through the protected path and state the subgradient convention at activation boundaries.

7.6 Metrics and decision gates

1. **Structural correctness.** For analytic prototypes, prove the fiber or kernel condition on the declared domain. For learned nonlinear models, report explicit witnesses or bounded-domain verification where tractable. Failure to find a collision is not proof of sufficiency.
2. **Target performance.** Report held-out normalized squared error for the original task and appropriate metrics for added tasks. The original 10^{-3} NMSE threshold can be retained for continuity, but it is not an exact-preservation criterion.
3. **Mechanism.** Audit the identified contrasts, negative common components, chart consistency, branch identities, anchors, and survival of records through the complete computation.
4. **Cost.** Report parameters, FLOPs, latency, bytes, and record count alongside accuracy. Attribute benefits to selectivity or gluing only after the appropriate controls.
5. **Reproducibility.** Preserve seeds, data splits, checkpoint selection rules, arithmetic settings, and per-case results. Report uncertainty across independent runs for learned comparisons.

Proceed to a larger model only if learned records repair the identified failures at a defensible budget and the gain survives ordinary-capacity, random-record, and unstructured-skip controls. If agreement

improves while target ambiguity persists, reject that representation. If a simple added channel explains the gain, report it without attributing the gain to atlas structure.

8 Established results and open questions

Selective preservation is mathematically possible: a representation may lose source distinctions while exactly retaining a declared target. The ReLU correction gives a constructive nonlinear example. The finite-pile identity shows that local corrections can compose. The linear theorems identify the missing directions requiring side records and give an explicit decoder. The atlas formulation separates compatibility, source explanation, and target observability.

The unresolved research question is whether a neural system can learn useful records and transport maps at a better accuracy–cost tradeoff than simpler baselines. Selection must occur before collapse, preserve required nonlinear branches, and remain sufficient for later computation. Upper-layer weights, successful prediction, small consistency loss, and full parameter-Jacobian rank do not remove those obligations.

The first implementation should be a small forward-boundary module with a declared record budget, a reconstruction interface to the atlas, and evaluation against the original certified failure types. Its acceptance statement is:

> Every ambiguity left by the complete retained state is invisible to the declared task family, and the implemented reconstruction exposes that information to the readout at the stated numerical precision.

9 Reproducible mathematical checks

The source bundle includes `verify_math.py` and its generated `verification_results.json`. The checks use exact integer and rational arithmetic for algebraic identities and explicit floating-point operations only for rounding counterexamples. All eight check groups passed.

Check	Domain and result
Two-input correction	289 pairs in $[-8, 8]^2$; absolute, contrast, gate, and reconstruction identities agree.
Signed coefficients	2,401 coefficient/input quadruples in $[-3, 3]^4$; general correction agrees with the target.
Positive position weights	294 ordered weighted cases; positive-homogeneity and weighted correction checks pass.
Original four-feature teacher	2,401 feature tuples in $[-3, 3]^4$; the pair corrections recover the target exactly.
Finite piles	3,125 five-leaf stacks in $[-2, 2]^5$, each through all 14 ordered binary groupings: 43,750 tree evaluations.
Deliberate ambiguities	CReLU after a merge, clipped sum plus contrast, and pairwise-product parity retain their target-changing collisions.
Selective linear record	625 inputs in $[-2, 2]^4$; one added coordinate repairs the required target while an irrelevant coordinate stays unrecoverable.
Finite precision	float32/64 rounding collisions and subtractive cancellation are reproduced; the stable pair formula is checked.

These finite checks validate calculations and witnesses. The proofs establish the general statements under their assumptions. No new network training, measured speedup, or accuracy improvement is reported.

A Derivations connecting the projection audit to overlap preservation

This appendix separates three objects: the historical training intervention, the information in a fixed linear measurement, and the information remaining after that measurement is rectified. The Gaussian calculation supplies an exact diagnostic for the original synthetic task. Its application to a selective overlap channel requires an additional conditioning argument.

A.1 Historical capacity, information loss, and anchor geometry

The archived student was a bias-free network

$$f_\theta(x) = \sum_{i=1}^{H} a_i \,\mathrm{ReLU}(w_i^\top x), \qquad W = [w_1, \dots, w_H] \in \mathbb{R}^{8\times H}. \tag{46}$$

Its teacher used four orthonormal directions:

$$f_*(x) = \sum_{j=1}^{4} c_j \,\mathrm{ReLU}(q_j^\top x), \qquad c = (-\tfrac{1}{2}, +\tfrac{1}{2}, -\tfrac{1}{2}, +\tfrac{1}{2}), \qquad q_i^\top q_j = \delta_{ij}. \tag{47}$$

Thus width four had enough capacity to express the scalar target. Nevertheless, every 8-to-4 linear map satisfies

$$\dim\ker W^\top = 8 - \operatorname{rank} W \geq 4. \tag{48}$$

For any $v \in \ker W^\top$, the inputs x and $x + v$ have identical preactivations and identical post-ReLU states. Preserving every input was therefore impossible in this architecture. Preserving the target was possible because it depends only on the four-dimensional teacher span. A successful representation could also discard distinctions within that span whenever they leave the target unchanged. The audit concerned collisions with *different* required targets, rather than every collision or an ordinary rank reduction [1, 2].

The historical QDO intervention acted on parameters. On the domain $r_i = \|w_i\| > 0$, define

$$u_i = w_i/r_i, \qquad b_i = a_i r_i, \qquad \phi(\theta) = \mathrm{pack}(u_1, \dots, u_H, b_1, \dots, b_H). \tag{49}$$

For positive s_i, replacing (w_i, a_i) by $(s_i w_i, a_i/s_i)$ leaves both f_θ and $\phi(\theta)$ unchanged, by positive homogeneity of ReLU. The coordinates intentionally remove this parameter ambiguity; they are not an encoding from which every original parameter value can be reconstructed.

With fixed historical parameters α_j, an orthonormal-column matrix $Q \in \mathbb{R}^{N\times k}$, and bandwidth $\tau > 0$, write

$$z = Q^\top \phi(\theta), \qquad z_j = Q^\top \phi(\alpha_j), \qquad \delta_j = z - z_j, \tag{50}$$

$$\mathcal{R}(\theta) = \sum_j \omega_j, \qquad \omega_j = \exp\left(-\frac{\|\delta_j\|^2}{2\tau^2}\right), \qquad r_z = \frac{1}{\tau^2}\sum_j \omega_j \delta_j. \tag{51}$$

The negative gradient is

$$F = -\nabla_\theta \mathcal{R} = D\phi(\theta)^\top Q r_z. \tag{52}$$

For $v = Qr_z$, unpacked into v_{u_i}, v_{b_i}, differentiation gives

$$\mathrm{d}u_i = r_i^{-1}(I - u_i u_i^\top)\,\mathrm{d}w_i, \qquad \mathrm{d}b_i = a_i u_i^\top \mathrm{d}w_i + r_i\,\mathrm{d}a_i, \tag{53}$$

$$F_{w_i} = r_i^{-1}(I - u_i u_i^\top) v_{u_i} + a_i v_{b_i} u_i, \qquad F_{a_i} = r_i v_{b_i}. \tag{54}$$

Consequently $w_i^\top F_{w_i} - a_i F_{a_i} = 0$: this auxiliary force is orthogonal to the infinitesimal positive-rescaling direction. Penalty invariance does not establish invariance of the complete Adam trajectory, nor does it establish injectivity of a forward representation [1]. The optimizer projection $Q^\top \phi(\theta)$ and input measurement $W^\top x$ act on different spaces. The proposal leaves this optimizer intervention behind and places its preservation requirement on the forward state and its overlap records.

This distinction also separates their possible guarantees. An auxiliary parameter force can change which representation is learned; it does not give a decoder access to information missing from a particular frozen forward state. An overlap record instead changes what that decoder receives. Its correctness must be established by a reconstruction or target-factorization identity for the retained observations. An optimizer result cannot substitute for that identity, even when both interventions were motivated by the same feature-merging failure.

A.2 The Gaussian ReLU kernel from polar coordinates

Let (U, V) be jointly standard normal with correlation $\rho \in [-1, 1]$. Set $\vartheta = \arccos \rho$ and express

$$U = Z_1, \qquad V = \cos\vartheta\, Z_1 + \sin\vartheta\, Z_2, \tag{55}$$

where Z_1, Z_2 are independent standard normals. Their polar representation is $(Z_1, Z_2) = R(\cos\varphi, \sin\varphi)$, with φ uniform on $[0, 2\pi)$, independent of R, and $\mathbb{E}R^2 = 2$. Therefore $U = R\cos\varphi$ and $V = R\cos(\varphi - \vartheta)$. The angles where both factors are positive form an interval equivalent to $[\vartheta - \pi/2, \pi/2]$. It follows that

$$\mathbb{E}[\mathrm{ReLU}(U)\,\mathrm{ReLU}(V)] = \frac{1}{\pi}\int_{\vartheta-\pi/2}^{\pi/2} \cos\varphi\cos(\varphi-\vartheta)\,\mathrm{d}\varphi \tag{56}$$

$$= \frac{\sin\vartheta + (\pi - \vartheta)\cos\vartheta}{2\pi}. \tag{57}$$

The second equality uses $2\cos\varphi\cos(\varphi-\vartheta) = \cos\vartheta + \cos(2\varphi-\vartheta)$. Thus the kernel used in the projection audit is

$$\boxed{\kappa(\rho) = \frac{\sqrt{1-\rho^2} + (\pi - \arccos\rho)\rho}{2\pi}.} \tag{58}$$

The endpoint checks are $\kappa(1) = 1/2$, $\kappa(0) = 1/(2\pi)$, and $\kappa(-1) = 0$. For nonzero vectors a, b and $X \sim \mathcal{N}(0, I_d)$, positive homogeneity yields

$$\mathbb{E}[\mathrm{ReLU}(a^\top X)\,\mathrm{ReLU}(b^\top X)] = \|a\|\,\|b\|\,\kappa\!\left(\frac{a^\top b}{\|a\|\,\|b\|}\right). \tag{59}$$

A zero vector contributes zero and requires no normalized correlation.

A.3 Teacher moments and normalization

For a standard normal Z, direct integration and symmetry give

$$\mathbb{E}\,\mathrm{ReLU}(Z) = \frac{1}{\sqrt{2\pi}}, \qquad \mathbb{E}\,\mathrm{ReLU}(Z)^2 = \frac{1}{2}, \qquad \mathrm{Var}(\mathrm{ReLU}(Z)) = \frac{1}{2} - \frac{1}{2\pi}. \tag{60}$$

The four preactivations in (47) are independent standard normals. Since $\sum_j c_j = 0$ and $\sum_j c_j^2 = 1$, the target $Y = f_*(X)$ satisfies

$$\mathbb{E}Y = 0, \qquad V_* := \mathrm{Var}(Y) = \frac{1}{2} - \frac{1}{2\pi}. \tag{61}$$

The historical normalized mean squared error divides by this V_* [1, 2]. This normalization belongs to that teacher; changing the target or input distribution requires recomputing it. The kernel calculation below does not require a zero target mean, because it subtracts second moments with identical means.

The positive and negative coefficients remain inside the double sum in the general risk formula. Cross terms can have either sign, even though the final conditional variance is nonnegative. Consequently an individual summand is not a nonnegative amount of information assigned to one feature. The calculation evaluates the complete target, including cancellations between its terms. This is also why replacing the target by a visually similar sum with different coefficients can change which contrasts must survive.

A.4 An exact fixed-projection prediction floor

Let $P = P^\top = P^2$ be an orthogonal projector and let $Y = f_*(X)$ have finite second moment. Define

$$m_P(PX) = \mathbb{E}[Y \mid PX], \qquad R(P) = \mathbb{E}[\mathrm{Var}(Y \mid PX)]. \tag{62}$$

For every square-integrable readout $\widehat{Y} = g(PX)$,

$$\mathbb{E}[(Y - \widehat{Y})^2] = R(P) + \mathbb{E}[(m_P(PX) - \widehat{Y})^2]. \tag{63}$$

Indeed, expand $Y - \widehat{Y} = (Y - m_P) + (m_P - \widehat{Y})$ and condition the cross term on PX. It vanishes because $\mathbb{E}[Y - m_P \mid PX] = 0$. Thus $R(P)$ is the optimal risk over *all* measurable readouts of the fixed projection. A restricted or unsuccessfully trained decoder can incur additional error. Changing the projection changes the floor.

For $P = P_W$, the projector onto $\operatorname{im} W$, observing $W^\top X$ is equivalent to observing $P_W X$: one direction follows from $W^\top P_W = W^\top$, and the other from $P_W X = (W^\top)^+ W^\top X$. The superscript $+$ denotes the Moore–Penrose pseudoinverse. This explains why the floor diagnoses the information available before ReLU [2].

The equivalence holds in exact arithmetic, regardless of the coordinates chosen within the observed subspace. An invertible change of measurement coordinates leaves the floor unchanged. In contrast, two projections of the same rank can have different floors because they observe different input directions. Ordinary rank therefore cannot distinguish all the successful and failed task representations. Numerical conditioning affects the stability of a reconstruction and must be evaluated separately from this exact information comparison.

Proposition A.1 (Gaussian fixed-projection formula)**.** *Suppose* $X \sim \mathcal{N}(0, I_d)$ *and* $Y = \sum_i c_i \operatorname{ReLU}(q_i^\top X)$ *with unit-norm* q_i*. Then*

$$\boxed{R(P) = \sum_{i,j} c_i c_j \left[\kappa(q_i^\top q_j) - \kappa(q_i^\top P q_j)\right].} \tag{64}$$

Proof. Take independent standard Gaussian vectors Z, E_1, E_2 and form

$$X_1 = PZ + (I - P)E_1, \qquad X_2 = PZ + (I - P)E_2. \tag{65}$$

Each copy has covariance $P + (I - P) = I$; their cross-covariance is P. Conditioned on the shared projection PZ, they have independent residuals and the correct conditional distribution of X. Therefore

$$\mathbb{E}[f_*(X_1) f_*(X_2)] = \mathbb{E}[m_P(PX)^2]. \tag{66}$$

The pair $(q_i^\top X_1, q_j^\top X_2)$ has unit marginal variances and correlation $q_i^\top P q_j$. Applying (58) term by term gives

$$\mathbb{E}[m_P(PX)^2] = \sum_{i,j} c_i c_j \kappa(q_i^\top P q_j), \tag{67}$$

$$\mathbb{E}[Y^2] = \sum_{i,j} c_i c_j \kappa(q_i^\top q_j). \tag{68}$$

Finally $R(P) = \mathbb{E}[Y^2] - \mathbb{E}[m_P(PX)^2]$, proving the formula. □

For non-unit teacher directions, let $n_i = \|q_i\| > 0$. The extension is

$$R(P) = \sum_{i,j} c_i c_j n_i n_j \left[\kappa\left(\frac{q_i^\top q_j}{n_i n_j} \right) - \kappa\left(\frac{q_i^\top P q_j}{n_i n_j} \right) \right]. \tag{69}$$

Both denominators use the *original* norms. The conditional copies keep their original marginal variances. Substituting projected norms would compute the ReLU kernel of Pq_i, Pq_j, a different quantity.

An independent expression for the optimal readout is also available. With $\mu_i = q_i^\top PX$ and $\sigma_i^2 = \|q_i\|^2 - \|Pq_i\|^2$, conditional Gaussianity gives $q_i^\top X \mid PX \sim \mathcal{N}(\mu_i, \sigma_i^2)$. Hence

$$m_P(PX) = \sum_i c_i \left[\sigma_i \varphi\left(\frac{\mu_i}{\sigma_i} \right) + \mu_i \Phi\left(\frac{\mu_i}{\sigma_i} \right) \right], \tag{70}$$

where φ, Φ are the standard normal density and distribution function. A term with $\sigma_i = 0$ is interpreted as $c_i \operatorname{ReLU}(\mu_i)$. This follows by integrating $(\mu_i + \sigma_i z)$ over $z > -\mu_i/\sigma_i$.

Proposition A.2 (The required span for the original teacher)**.** *For the orthonormal teacher in* (47) *and standard Gaussian inputs,* $R(P) = 0$ *if and only if every* q_j *belongs to* $\operatorname{im} P$.

Proof. If all teacher directions are observed, each preactivation and hence the target is a function of PX, proving sufficiency. Conversely, zero risk means the target is almost surely constant on almost every conditional fiber. The conditional Gaussian has full support on that fiber. Continuity of the ReLU teacher therefore makes it constant everywhere on each such fiber. Those observed projections form a dense set; continuity extends this invariance to every fiber. Thus for every $v \in \ker P$,

$$f_*(x + tv) = f_*(x) \qquad \text{for every } x \text{ and } t. \tag{71}$$

Orthonormality permits a nonempty open region where exactly teacher feature j is positive. There the gradient is $c_j q_j$, so (71) implies $c_j q_j^\top v = 0$. Every coefficient is nonzero, hence q_j is orthogonal to $\ker P$ and belongs to $\operatorname{im} P$. □

This result is specific to this teacher with nonzero coefficients and independently variable orthonormal features. It identifies the four required directions without demanding preservation of the other four input directions. It also concerns the unrectified observation: preserving that span alone does not prove that a later nonlinear representation remains sufficient, or that training learns the necessary decoder.

A.5 CReLU gives an exact floor; a selective hybrid gives a bound

CReLU retains both rectified phases of a measured vector:

$$H_c = \mathrm{CReLU}(W^\top X) = \big(\mathrm{ReLU}(W^\top X), \mathrm{ReLU}(-W^\top X)\big). \tag{72}$$

Subtraction recovers $W^\top X$ exactly. Conversely, H_c is a deterministic function of that measurement. Consequently conditioning on H_c, $W^\top X$, or $P_W X$ supplies identical information, and

$$\inf_g \mathbb{E}[(Y - g(H_c))^2] = R(P_W). \tag{73}$$

This is the measurement-preservation property behind the original CReLU filter-span reconstruction proposition, without max-pooling [4]. It does not restore any component in $\ker W^\top$.

Now consider the selective hybrid

$$H = \big(\mathrm{ReLU}(W^\top X), \mathrm{CReLU}(U^\top X)\big), \qquad M = [W\ U], \qquad T = P_M X. \tag{74}$$

Here H is a function of T, but the reverse need not hold. Set $m_M(T) = \mathbb{E}[Y \mid T]$ and $R_H = \mathbb{E}[\mathrm{Var}(Y \mid H)]$. The conditional variance decomposition, using that H is determined by T, gives

$$\begin{aligned} R_H &= R(P_M) + \mathbb{E}[\mathrm{Var}(m_M(T) \mid H)] \\ &= R(P_M) + \mathbb{E}[(m_M(T) - \mathbb{E}[Y \mid H])^2] \ \geq\ R(P_M). \end{aligned} \tag{75}$$

For completeness, write $Y - \mathbb{E}[Y \mid H] = (Y - m_M) + (m_M - \mathbb{E}[Y \mid H])$. The cross term again vanishes on conditioning on T, and the tower property gives $\mathbb{E}[m_M \mid H] = \mathbb{E}[Y \mid H]$. This proves the exact gap formula.

Equality holds precisely when $m_M(T)$ is recoverable from H almost surely. Recovering all of T is sufficient but stronger than necessary. Merely making $\operatorname{im} M$ contain the teacher span proves no such equality: the unrectified measurements can contain a distinction that the main ReLU branch still deletes.

Accordingly, a positive augmented-span floor rules out exact target recovery from the hybrid. A zero augmented-span floor leaves the second term of (75) to be checked. This term measures the target-relevant ambiguity introduced by the actual retained observation, after all signed measurements that were available before rectification have been taken into account. It gives a precise test of whether a proposed selective side channel supplies enough information.

An exact Gaussian example demonstrates a strict gap. Let $X \sim \mathcal{N}(0, I_2)$, $W = e_1$, $U = e_2$, and $Y = \mathrm{ReLU}(-X_1)$. Then $M = I_2$ and $R(P_M) = 0$. The side channel reveals X_2, which is independent of Y. On $X_1 > 0$ the target is zero; on $X_1 \leq 0$ the main observation is always zero and the conditional target is half-normal. Therefore

$$R_H = \tfrac{1}{2}\,\mathrm{Var}(-X_1 \mid X_1 \leq 0) = \frac{1}{2} - \frac{1}{\pi} > 0. \tag{76}$$

Thus even complete augmented linear coverage can coexist with strictly positive optimal prediction error after selective rectification.

A.6 Scope of the diagnostic

The conditional-risk identities hold for any square-integrable target. The closed form (64) additionally requires Gaussian inputs, a fixed linear observation, and a ReLU teacher of the stated form. Arbitrary

hidden states, nonlinear transition maps, pooling, quantization, or a different input law do not automatically satisfy those assumptions. The algebraic CReLU inverse also presumes that both measured phases remain available without additional destructive processing.

Training the matrices first and then freezing them is compatible with the fixed-projection calculation. Choosing a different measurement matrix as a function of the current input is a different observation model: the selection rule and any retained selector must be included in the conditioning information. Substituting an input-dependent projector into the fixed-matrix formula is not justified by the derivation above. Such adaptive records need their own sufficiency argument or conditional-risk calculation.

Finally, $R_H = 0$ states target sufficiency almost surely under the chosen distribution. A guarantee for every admissible input requires a pointwise factorization through the retained representation. Neither a successful probe nor failure to find a collision supplies that proof. These distinctions let the projection audit support the proposal as a precise diagnostic without turning it into a general information-preservation theorem.

B Grid overlaps, rotations, and full-reconstruction limits

These calculations arose during the discussion. They delimit what weighted overlap alone can guarantee and explain why full serialization is not the main proposal.

B.1 Independent real-valued grids

Flatten each grid to $X_\ell \in \mathbb{R}^{81}$. For

$$Y = \sum_{\ell=0}^{K-1} w_\ell X_\ell,$$

the measurement matrix is

$$M = [\, w_0 I_{81} \;\cdots\; w_{K-1} I_{81} \,].$$

If at least one weight is nonzero, $\operatorname{rank} M = 81$, hence $\dim \ker M = 81(K-1)$. For two nonzero weights, explicit collisions are

$$X_i' = X_i + w_j T, \qquad X_j' = X_j - w_i T.$$

The changes cancel in Y. A sufficiently small nonzero T also gives collisions in the interior of bounded value ranges. Different weights cannot remove this linear ambiguity. A linear representation of K independent continuous grids needs at least K scalar-grid output channels.

B.2 A weighted shadow with retained contrasts

For positive weights, retain

$$S = \sum_{\ell=0}^{K-1} w_\ell X_\ell, \qquad D_\ell = X_\ell - X_0, \quad 1 \le \ell < K.$$

Let $W = \sum_\ell w_\ell$. Substituting $X_\ell = X_0 + D_\ell$ gives

$$S = W X_0 + \sum_{\ell=1}^{K-1} w_\ell D_\ell,$$

so the inverse is

$$X_0 = \frac{S - \sum_{\ell=1}^{K-1} w_\ell D_\ell}{W}, \qquad X_\ell = X_0 + D_\ell.$$

There are K channels, and the signed contrasts must not be clipped.

A well-conditioned alternative is $y = Qx$ at each cell, where Q is orthogonal and its first row is $w^\top / \|w\|_2$. Then

$$x = Q^\top y, \qquad \|Qx\|_2 = \|x\|_2, \qquad \kappa_2(Q) = 1.$$

For $a = w/\|w\|_2 \neq e_0$, an explicit completion is

$$v = e_0 - a, \qquad Q = I - \frac{2vv^\top}{v^\top v}.$$

Direct multiplication gives $Q^\top = Q$, $Q^2 = I$, and $Qe_0 = a$, so its first row is $a^\top$. If $a = e_0$, use $Q = I$. These are exact-real constructions, not bitwise floating-point guarantees.

B.3 Known quarter-turn copies of one source

Let R be the exact 90° rotation permutation of the grid, so $R^4 = I$. For ordered layers X, RX, R^2X, R^3X and increasing weights $1, 2, 4, 8$, set

$$Y = \frac{X + 2RX + 4R^2X + 8R^3X}{15}. \tag{77}$$

Multiplication yields

$$(2R - I)(I + 2R + 4R^2 + 8R^3) = 16R^4 - I = 15I,$$

and therefore

$$\boxed{X = 2RY - Y.} \tag{78}$$

The descending convention $(8X + 4RX + 2R^2X + R^3X)/15$ instead has inverse $X = 2Y - RY$. Coefficient order and inverse must remain paired.

For any $r > 1$,

$$T_r = \frac{I + rR + r^2R^2 + r^3R^3}{1 + r + r^2 + r^3}, \qquad T_r^{-1} = \frac{rR - I}{r - 1}.$$

Indeed, $(rR - I)(I + rR + r^2R^2 + r^3R^3) = (r^4 - 1)I$, and $r^4 - 1 = (r - 1)(1 + r + r^2 + r^3)$.

A 9×9 grid has one fixed center and twenty four-cell orbits. The rotation eigenvalues $1, -1, i, -i$ therefore have multiplicities $21, 20, 20, 20$. Since R is an orthogonal permutation, a polynomial in R is normal, and its singular values are the magnitudes of its eigenvalues. At $r = 2$ they are

$$1 \text{ (21 times)}, \qquad 1/\sqrt{5} \text{ (40 times)}, \qquad 1/3 \text{ (20 times)}.$$

Thus the condition number is 3, and $\|(2R - I)\varepsilon\|_2 \leq 3\|\varepsilon\|_2$. The same factor bounds maximum cell error by the triangle inequality. More generally, $\kappa_2(T_r) = (r + 1)/(r - 1)$, because the largest and smallest inverse magnitudes occur at the rotation eigenvalues -1 and 1.

Equal averaging $(I + R + R^2 + R^3)/4$ retains only one average per orbit, so it has rank 21 and nullity 60. For arbitrary four weights, the map is invertible on this grid exactly when

$$p(t) = w_0 + w_1 t + w_2 t^2 + w_3 t^3$$

is nonzero at $1, -1, i, -i$. Strictly increasing positive weights satisfy this: $p(1) > 0$, $p(-1) = (w_0 - w_1) + (w_2 - w_3) < 0$, and

$$p(i) = (w_0 - w_2) + i(w_1 - w_3) \neq 0.$$

They can still be poorly conditioned as they approach equal weighting.

For original eight-bit cells, store the unnormalized integer

$$N = X + 2RX + 4R^2X + 8R^3X.$$

Its cells lie in $[0, 3825]$, so 12 bits suffice per output cell. The inverse $X = (2RN - N)/15$ is exact on valid codes with sufficiently wide integer intermediates; each numerator is divisible by 15.

There is only one source grid's worth of independent unknowns here. The result cannot encode four independent real-valued grids into one. Quarter-turn permutations, known orientation convention, and known transform order are assumptions. Arbitrary-angle interpolation or cropping needs separate analysis. The map is rotation-equivariant, $T_r R = RT_r$, rather than rotation-invariant. Discarding orientation while asking to recover it would require additional pose information.

B.4 Exact positional packing for a finite alphabet

For original b-bit unsigned values, let $q = 2^b$ and $x_\ell \in \{0, \ldots, q-1\}$. Define

$$C = \sum_{\ell=0}^{K-1} q^\ell x_\ell, \qquad x_\ell = \left\lfloor C/q^\ell \right\rfloor \bmod q. \tag{79}$$

The contribution from the highest differing position j differs by at least q^j. The greatest possible opposing lower contribution is

$$(q-1)\sum_{i<j} q^i = q^j - 1.$$

It cannot cancel the higher difference. Division and reduction modulo q then give the stated inverse.

For arbitrary scalar weights, injectivity on this alphabet is equivalent to

$$\sum_\ell w_\ell \delta_\ell \neq 0 \quad \text{for every nonzero integer } \delta \text{ with } |\delta_\ell| \leq q-1.$$

A collision gives such a difference vector. Conversely, every such difference is realized by $x_\ell = \max(\delta_\ell, 0)$ and $y_\ell = \max(-\delta_\ell, 0)$. A sufficient dominance condition is

$$w_j > (q-1)\sum_{i<j} w_i, \qquad w_0 > 0.$$

The highest-position argument proves it. It is not necessary: binary weights $2, 3, 4$ have eight distinct subset sums even though the last weight does not exceed the sum of the earlier two.

Represent a variable-depth ordered stack by (C, K). Placing stack B above A gives

$$(C_A, K_A) \star (C_B, K_B) = (C_A + q^{K_A} C_B,\ K_A + K_B).$$

Both groupings of three stacks have code

$$C_A + q^{K_A} C_B + q^{K_A + K_B} C_C,$$

which proves associativity. Given the lower split depth, reduction modulo q^{K_A} and integer division recover the two blocks. Appending a new upper layer uses $C_{\text{new}} = C + q^K x_{\text{new}}$; the update $qC + x_{\text{new}}$ instead makes older layers heavier.

Depth must be retained because a zero upper layer changes no value code. Grouping boundaries and source identities require metadata if they count as information. Values outside the digit range create carries: $(q, 0)$ and $(0, 1)$ encode identically. Known unequal digit ranges admit mixed-radix weights $w_\ell = \prod_{j<\ell} q_j$, with their bases available to the decoder.

B.5 Capacity and brightness precision

There are $q^K = 2^{Kb}$ independent cell stacks. A fixed-length lossless code therefore needs at least Kb bits per cell, or $81Kb$ bits for the grid, excluding metadata. A single spatial grid can carry the stack only by increasing precision or retaining additional state. Positional packing meets this payload count exactly.

For $K \geq 1$, normalize the integer code to $z = C/(q^K - 1)$, and write $M = q^K - 1$. Adjacent valid brightness levels are separated by $\Delta = 1/M$. With exact scaling and rounding, decoding by

$$\widehat{C} = \text{round}(M\widehat{z})$$

is guaranteed when

$$|\widehat{z} - z| < \frac{1}{2M}.$$

If the implemented scaled value is $M\widehat{z} + \delta$, a sufficient condition including that arithmetic error is

$$M|\widehat{z} - z| + |\delta| < \frac{1}{2}.$$

For four eight-bit layers and exact decoding arithmetic, the normalized error threshold is about 1.16415×10^{-10}. For any N distinct scalar codes in $[0, 1]$, the $N - 1$ gaps sum to at most one, so the minimum gap is at most $1/(N - 1)$. Alternative weights cannot remove that fixed-range robustness limit.

This is a full-reconstruction reference and a capacity check. Quantizing original floating-point values into new eight-bit shades is not lossless. Preserving original floating-point bit patterns requires their full stored bits, including sign and exponent. Such packing does not automatically give a useful numerical representation for learning.

C ReLU, reversible computation, and numerical exactness

For real scalars, $(\rho(x), \rho(-x))$ recovers x. A related bound applies to one bias-free layer $h(x) = \rho(Ax)$ on all of $\mathbb{R}^n$: global injectivity requires at least $2n$ output coordinates [7]. Ignore identically zero rows and choose a generic point away from the hyperplanes of the nonzero rows. Active rows must span n dimensions for local injectivity. At $-x$, the complementary nonzero rows must also span n. There must therefore be at least $2n$ rows. Pairing the identity and negative identity attains the bound. It does not impose the same requirement on every architecture, restricted domain, or target-specific encoding.

ReLU can occur inside a larger invertible operation. For example,

$$u = A + F(B), \qquad v = B + G(u)$$

has inverse

$$B = v - G(u), \qquad A = u - F(B).$$

The functions F, G themselves need not be invertible. Additive coupling is a standard construction in invertible networks [8]. Its full carried state retains the dimensions needed for the inverse; it does not compress arbitrary new independent inputs into insufficient capacity.

Algebraic reversibility is distinct from bitwise reversibility. IEEE float32 has 24 significand bits and float64 has 53 [10]. With ordinary round-to-nearest,

$$\text{float32}(2^{24} + 1) = \text{float32}(2^{24}),$$

and the analogous binary64 collision occurs at $2^{53} + 1$. A floating-point additive coupling can therefore map different inputs to the same stored output even though its real-arithmetic formula has an inverse.

For unnormalized integer-valued radix codes, all Kb-bit payloads are representable in float32 if $Kb \leq 24$, and in float64 if $Kb \leq 53$, with sufficient exponent range and exact intermediate operations. Correctly implemented unsigned Kb-bit integer storage retains the entire payload. Original float32 payloads require 32 stored bits each, not merely their 24-bit significands.

For fixed-width integer states and deterministic integer-valued maps F, G on the appropriate finite state spaces, modular coupling

$$u = (A + F(B)) \bmod M, \qquad v = (B + G(u)) \bmod M$$

has an exact inverse by modular subtraction. Integer bijective flows provide a related discrete approach [9]. Wraparound changes the monotone darkness interpretation, and training a discrete transformation needs its own analysis.

Report three different claims separately: algebraic target sufficiency, stability at a stated error level, and bitwise recovery when actually required. The selective architecture is not granted the last claim by the first.

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